

AARYA RAY CHAUDHURI

AI / Machine Learning Engineer · Applied Computer Science (AI Specialization), Thomas More

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Applied Computer Science student specializing in AI who builds things that leave the notebook. Right now I'm modeling executive writing styles with stylometry and sentence-transformer embeddings at a US startup, at **78% author-identification accuracy on production data**. Before that: an offline IoT system sending East African farmers crop advice over SMS, and full-stack Laravel apps for real companies. I learn by getting my hands dirty: Python and PyTorch on one side, Docker and a Linux homelab on the other. Looking for AI/ML internships where the model actually has to run somewhere.

WORK EXPERIENCE

- Twelfth Brain · Research Intern, Machine LearningJune 2026 – August 2026
California, United States
 - Building the machine learning foundation of **Voice Fingerprint**, a system that models an individual executive's writing style: stylometric feature extraction with **spaCy** and **NLTK**, combined with sentence-transformer embeddings (**SBERT**, **PyTorch**).
 - Own the feature extraction, voice embedding, and benchmarking pipelines end to end in Python, currently reaching **78% author-identification accuracy on production data**. It is real customer text, not a clean dataset, so most of the work goes into deciding which stylistic signals carry an author at all.
 - Helped the other marketing interns set up an **agentic pipeline** for post ideation and creation across Instagram, LinkedIn, Substack and X, and produced Instagram posts for my mentor's accounts.
- AllLabs · Machine Learning InternJune 2023 – August 2024
Kolkata, India · Two consecutive summers, invited back after the first
 - Built and evaluated models in Python (**Pandas**, **scikit-learn**, **Jupyter**) across the full workflow, from data cleaning to validation. Got a diabetes prediction model to **93% accuracy** by comparing regression approaches and weighing hyperparameter tuning against better dataset cleaning.
 - Second summer, moved from classical ML into deep learning: built classifiers in both **PyTorch** and **Keras**, and evaluated them properly with precision/recall and cross-validation, not just accuracy.
- Client Projects, Thomas More · Full-Stack Developer (student team)September 2025 – June 2026
 - Built and delivered web applications for two real companies, **Grow Kasterlee** and **The Andersons**, on the **TALL stack** (Tailwind, Alpine.js, **Laravel**, Livewire): a booking platform for a gym and a resident management system for a care home.
 - Worked in a student team, taking both from client requirements through **UML** data modeling to build and handover.

PROJECTS

- Scout · Open-source search layer for AI agents2026 – Ongoing
github.com/aaryaray1/scout-agent-search
 - Started this after realizing web data is formatted for humans, not machines: Scout gives **AI agents** a retrieval interface built for machine consumption instead of scraped HTML.
 - Written entirely in **Python**. Still in active development, with progress posted on my GitHub and LinkedIn.
- Captain Alex · Offline IoT and ML for East African farmersMarch 2026
github.com/PJUytterhoeven/CaptainAlex
 - Built an offline system that reaches farmers over plain **SMS** where there is no reliable internet: soil readings and live weather in, daily crop advice out.
 - Trained the **Random Forest** yield model on satellite and weather data, delivered advice in English and Swahili, and ran the stack on **Armbian Linux** on low-power hardware.
- Speeds7ers · Tron-style game with a self-taught AI opponentFebruary 2025
github.com/aaryaray1/speeds7ers
 - Built a **Deep Q-Learning** agent (**PyTorch/TensorFlow**) that studies your play style and adapts to beat you, instead of following a fixed strategy.
 - Wrote the game and training loop in Python/**PyGame**; tuning the reward function and exploration decay mattered more than any difficulty setting I could have written by hand.

TECHNICAL SKILLS

Languages	Python, Java, JavaScript, PHP, SQL, HTML/CSS
Machine Learning	PyTorch, TensorFlow, Keras, scikit-learn, Pandas, NumPy, Jupyter
ML Techniques	Deep learning, neural networks, reinforcement learning (Deep Q-Learning), random forests, regression, classification, feature engineering, hyperparameter tuning, cross-validation, model evaluation
NLP	Natural Language Processing (NLP), spaCy, NLTK, SBERT / sentence-transformers, text embeddings, stylometry, AI agent tooling and retrieval interfaces
Web & Data	Laravel, TALL stack (Tailwind, Alpine.js, Laravel, Livewire), REST APIs, Qlik, UML (StarUML)
Tools & Infra	Docker, Git, Linux (Armbian/Debian), PowerShell, edge deployment (NVIDIA Jetson Nano), plus a self-hosted Linux/Docker homelab for networking and deployment practice
Spoken	English (fluent), Hindi (fluent), Bengali (intermediate), French (intermediate)

EDUCATION

- Thomas More University of Applied Sciences · BAsc, Applied Computer Science (AI Specialization)September 2024 – September 2027
Previously: Delhi Public School International (India) · Oakdale High School (California, USA) · SHAPE International High School (Belgium)

CERTIFICATIONS & PROGRAMS

- NVIDIA Getting Started with AI on Jetson Nano · Omniverse Extension Development · Cisco AI/ML Development Program and Hackathon (2023)
- Forage J.P. Morgan (Quantitative Research) · J.P. Morgan (Software Engineering) · Hewlett Packard Enterprise (Software Engineering)